

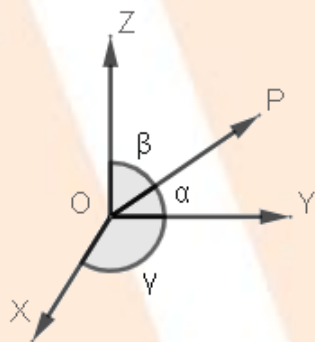
Revision Notes

Class 12 - Maths

Chapter 11 – Three Dimensional Geometry

1. Direction Cosines and Direction Ratios

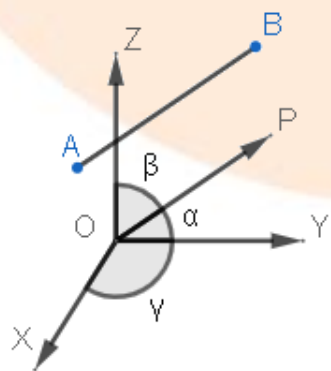
- (i) Consider a line OP passing through origin.



The angles the line OP makes with the x, y and z axes be α, β and γ .
Then, $\cos \alpha, \cos \beta$ and $\cos \gamma$ are the direction cosines of the line OP.
They are represented as:
 $l = \cos \alpha, m = \cos \beta, n = \cos \gamma$

- (ii) For lines not passing through the origin, the direction cosines are found using the direction ratios.
Consider line AB. Now draw line parallel to line AB passing through origin, i.e., OP.

Two parallel lines have the same direction cosines.



So, a, b, c will be proportional to the l, m, n then a, b, c are the Direction Ratios of the line, such that

$$a = kl, b = km, c = kn$$

(iii) If a, b, c are direction ratio of any line, then the vector parallel to this line will be $a\hat{i} + b\hat{j} + c\hat{k}$.

(iv) If l, m, n are direction cosines of any line, then the unit vector parallel to this line will be $l\hat{i} + m\hat{j} + n\hat{k}$.

2. Relation between the direction cosines and direction ratios of a line

(i) If l, m, n are direction cosines of a line, then

$$l^2 + m^2 + n^2 = 1$$

(ii) If l, m, n are direction cosines and a, b, c are direction ratios of a vector, then

$$l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

3. Direction cosines and Direction ratios of a line passing through two points

(i) Let two points be A and B such that their coordinates are (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively. Then the direction cosines of a line passing through AB are:

$$l = \frac{x_2 - x_1}{|AB|}, m = \frac{y_2 - y_1}{|AB|}, n = \frac{z_2 - z_1}{|AB|}$$

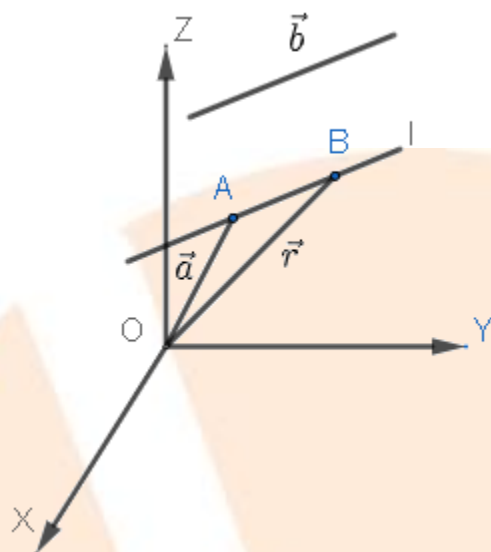
$$\text{Where } |AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

And the direction ratios of the line AB are:

$$a = x_2 - x_1, b = y_2 - y_1, c = z_2 - z_1$$

4. Equation of a line in space

(i) Equation of line through a given point and parallel to a vector:



The vector equation of line passing through point A with position vector $\vec{a}$ and parallel to vector $\vec{b}$ is given by:

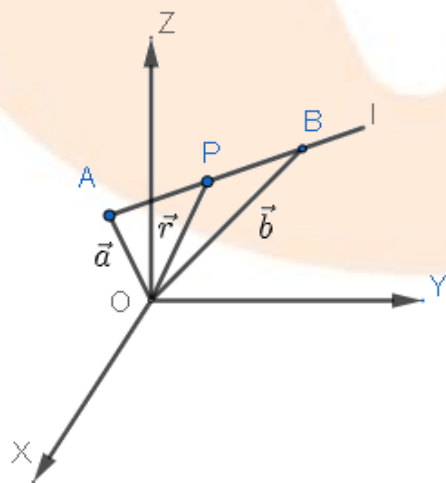
$$\vec{r} = \vec{a} + \lambda \vec{b}$$

The Cartesian equation of line passing through point A (x_1, y_1, z_1) with direction ratios a, b, c is given by:

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = r$$

The general point on this line will be given by: $(x_1 + ar, y_1 + br, z_1 + cr)$

(ii) Equation of line through a two given points:



The vector equation of line passing through point A with position vector $\vec{a}$ and point B with position vector $\vec{b}$ is given by:

$$\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$$

The Cartesian equation of line passing through point A (x_1, y_1, z_1) and B (x_2, y_2, z_2) is given by:

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

The general point on this line will be given by: $(x_1 + ar, y_1 + br, z_1 + cr)$

5. Angle between two line segments

- (i) The angle between two lines with direction ratios a_1, b_1, c_1 and a_2, b_2, c_2 is given by:

$$\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Here $a_1 \hat{i} + b_1 \hat{j} + c_1 \hat{k}$ and $a_2 \hat{i} + b_2 \hat{j} + c_2 \hat{k}$ are two vector parallel to the given two lines respectively.

- (ii) The two lines are perpendicular if $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$.

- (iii) The two lines are parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

6. Skew Lines

- (i) The lines which are not parallel nor lying in same plane, i.e., they are not intersecting are known as Skew Lines. The condition for skew lines is:

$$\Delta = \begin{vmatrix} \alpha' - \alpha & \beta' - \beta & \gamma' - \gamma \\ l & m & n \\ l' & m' & n' \end{vmatrix} \neq 0$$

(ii) The vector condition for two lines to be skew are:

$$(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_1 \times \vec{a}_2) \neq 0$$

7. Shortest Distance between Two Lines

(i) Distance between two skew lines, l_1 and l_2 with equations $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$ is given by:

$$d = \frac{|(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 \times \vec{a}_1)|}{|\vec{b}_1 \times \vec{b}_2|}$$

(ii) The Cartesian distance between two skew lines, l_1 and l_2 is given by:

$$d = \frac{\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1 c_2 - b_2 c_1)^2 + (c_1 a_2 - c_2 a_1)^2 + (a_1 b_2 - a_2 b_1)^2}}$$

Where $l_1 = \frac{x - x_1}{a_1} = \frac{y - y_1}{b_1} = \frac{z - z_1}{c_1}$, $l_2 = \frac{x - x_2}{a_2} = \frac{y - y_2}{b_2} = \frac{z - z_2}{c_2}$ are the skew lines.

(iii) Distance between two parallel lines, l_1 and l_2 with equations $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$ is given by:

$$d = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$$

8. Equation of a plane

(i) Vector form of equation of plane in normal form is given by:

$$\vec{r} \cdot \hat{n} = d$$

Where $\vec{r}$ is the position vector of the point on the given plane, $\hat{n}$ is the normal unit vector to the plane from origin and d is the length of the normal unit vector from the origin to the plane.

- (ii) Cartesian form of equation of plane in normal form is given by:

$$lx + my + nz = d$$

Where $P(x, y, z)$ is the any point on the given plane, $\hat{n}$ is the normal unit vector to the plane from origin and l, m, n are the direction cosines of the normal unit vector.

- (iii) If $\vec{r} \cdot (\hat{a}i + \hat{b}j + \hat{c}k) = d$ is the vector equation of the plane, then the Cartesian equation of the plane will be $(ax + by + cz) = d$. Here a, b, c are the direction ratios of the normal to the plane.

9. Equation of a plane perpendicular to a given vector and passing through a given point

- (i) The vector form of the equation of the plane is:

$$(\vec{r} - \vec{a}) \cdot \vec{N} = 0$$

Where $\vec{r}$ is the position vector of the any point on the given plane, $\vec{N}$ is the given vector to the plane from origin and $\vec{a}$ is the position vector of the given point on the given plane

- (ii) Cartesian form of equation of plane is given by:

$$A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$$

Where $P(x, y, z)$ is the any point on the given plane, (x_1, y_1, z_1) is the given point on the given plane and A, B, C are the direction ratios of the given vector.

10. Equation of a plane passing through three non collinear points

- (i) The vector form of the equation of the plane is:

$$(\vec{r} - \vec{a}) \cdot [(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})] = 0$$

Where $\vec{r}$ is the position vector of the any point on the given plane, $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of the three non collinear points on the plane.

- (ii) Cartesian form of equation of plane is given by:

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

Where $P(x, y, z)$ is the any point on the given plane, $(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3)$ are the three non collinear points on the plane.

11. Intercept form of the equation of a plane

- (i) The equation is:

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

Where a, b, c are the intercepts the plane makes with x, y, z axes, respectively.

- (ii) The plane intercepts x axis at $(a, 0, 0)$. Similarly, the plane intercepts y axis at $(0, b, 0)$ and the plane intercepts z axis at $(0, 0, c)$.

12. Plane passing through the intersection of two given planes

- (i) The vector form of equation of the plane passing through the intersection of the two given planes $\vec{r} \cdot \hat{n}_1 = d_1$ and $\vec{r} \cdot \hat{n}_2 = d_2$ is given by:
- $$\vec{r} \cdot (\vec{n}_1 + \lambda \vec{n}_2) = d_1 + \lambda d_2$$

- (ii) The Cartesian form of equation of the plane passing through the intersection of the two given planes $\vec{n}_1 = A_1\hat{i} + B_1\hat{j} + C_1\hat{k}$ and $\vec{n}_2 = A_2\hat{i} + B_2\hat{j} + C_2\hat{k}$ is given by:
- $$(A_1x + B_1y + C_1z - d_1) + \lambda(A_2x + B_2y + C_2z - d_2) = 0$$

13. Coplanar lines

- (i) Two lines are coplanar if:

$$\begin{vmatrix} \alpha - \alpha' & \beta - \beta' & \gamma - \gamma' \\ l & m & n \\ l' & m' & n' \end{vmatrix} = 0$$

Where $\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n}$ and $\frac{x - \alpha'}{l'} = \frac{y - \beta'}{m'} = \frac{z - \gamma'}{n'}$ are the two lines.

- (ii) The plane containing the coplanar lines is:

$$\begin{vmatrix} x - \alpha & y - \beta & z - \gamma \\ l & m & n \\ l' & m' & n' \end{vmatrix} = 0$$

- (iii) The condition for co-planarity in vector form:

$$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$$

Where $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \lambda \vec{b}_2$ are the given lines.

14. Angle between two planes

- (i) Angle between the two normal to the two planes is known as the angle between the two planes.

- (ii) The vector form of the angle between the two planes is given by:

$$\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|}$$

Where $\vec{n}_1, \vec{n}_2$ are the two normal to the two planes respectively.

- (iii) The Cartesian form of the angle between the two planes is given by:

$$\cos \theta = \left| \frac{A_1 A_2 + B_1 B_2 + C_1 C_2}{\sqrt{A_1^2 + B_1^2 + C_1^2} \sqrt{A_2^2 + B_2^2 + C_2^2}} \right|$$

Where A_1, B_1, C_1 and A_2, B_2, C_2 are the direction ratios of the normal to the given planes.

(iv) The planes are perpendicular to each other if $\vec{n}_1 \cdot \vec{n}_2 = 0$ or $A_1A_2 + B_1B_2 + C_1C_2 = 0$.

(v) The planes are parallel to each other if $\vec{n}_1$ is parallel to $\vec{n}_2$ or $\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$.

15.Distance of a Point from a Plane

(i) The vector form of the distance is:

$$|\vec{d} - \vec{a} \cdot \hat{n}|$$

Where $\vec{a}$ is the position vector of the given point and $\vec{r} \cdot \hat{n} = d$ is the equation of the plane.

(ii) The Cartesian form of the distance is:

$$\left| \frac{Ax_1 + By_1 + Cz_1 - D}{\sqrt{A^2 + B^2 + C^2}} \right|$$

Where $P(x_1, y_1, z_1)$ is the given point and $Ax + By + Cz = D$ is the Cartesian equation of the plane.

(iii) When $\vec{r} \cdot \vec{N} = d$ is the equation of the plane and $\vec{N}$ normal to the plane, then the perpendicular distance is given by:

$$\frac{|\vec{a} \cdot \vec{N} - d|}{|\vec{N}|}$$

(iv) The perpendicular distance from origin to the plane $\vec{r} \cdot \vec{N} = d$ is given by:

$$\frac{|d|}{|\vec{N}|}$$

16. Angle between a line and a plane

- (i) The complement of the angle between the normal to the plane and the line is known as the angle between the line and the plane.
- (ii) The vector form of the angle between the line and the plane is given by:

$$\sin(90 - \theta) = \cos \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|}$$

Where $\vec{r} = \vec{a} + \lambda \vec{b}$ is the equation of the line and $\vec{r} \cdot \vec{n} = d$ is the equation of the plane.